

C Programming – Summer Training

Assignment 4 Submission

Programs 1 – 10

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Project 1 — Task 4.1

WAP in C to find the largest and smallest element in an array

Source Code

```
// Task 4.1 WAP in C to find the largest and smallest element in array
```

```
#include<stdio.h>

int main(){
    int arr1[5];
    int i, largest, smallest;

    printf("Enter 5 elements: ");
    for(i = 0; i < 5; i++){
        scanf("%d", &arr1[i]);
    }

    largest = arr1[0];
    smallest = arr1[0];

    for(i = 1; i < 5; i++){
        if(arr1[i] > largest){
            largest = arr1[i];
        }
        if(arr1[i] < smallest){
            smallest = arr1[i];
        }
    }

    printf("Largest element : %d\n", largest);
    printf("Smallest element : %d\n", smallest);

    return 0;
}
```


Project 2 — Task 4.2

WAP in C to find the sum of array elements

Source Code

```
// Task 4.2 WAP in C to find sum of array elements
```

```
#include<stdio.h>
```

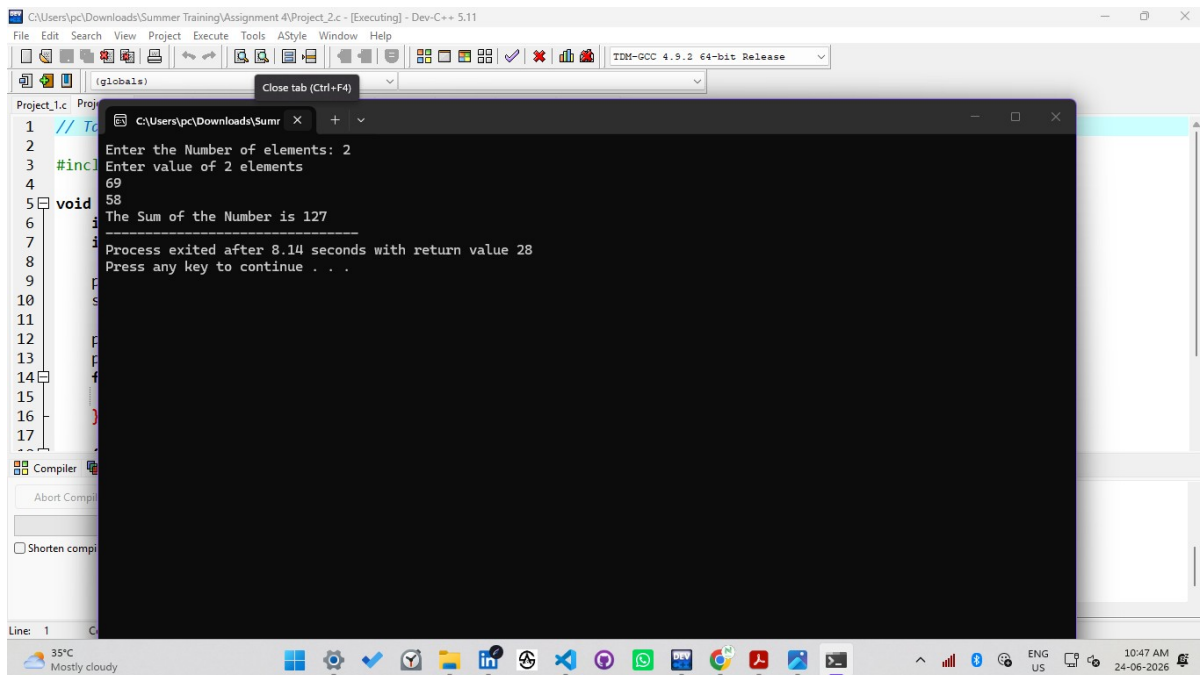
```
void main(){
    int arr1[100];
    int i,n,sum=0;

    printf("Enter the Number of elements: ");
    scanf("%d",&n);

    printf("Enter value of %d elements ",n);
    printf("\n");
    for(i=1 ; i<=n;i++){
        scanf("%d",&arr1[i]);
    }

    for(i=1 ; i<=n ; i++){
        sum += arr1[i];
    }
    printf("The Sum of the Number is %d",sum);
}
```

Output



```
C:\Users\pc\Downloads\Summer Training\Assignment 4\Project_2.c - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
(globals) Close tab (Ctrl+F4)
Project_1.c Project_2.c
1 // Task 4.2 WAP in C to find sum of array elements
2
3 #include<stdio.h>
4
5 void main(){
6     int arr1[100];
7     int i,n,sum=0;
8
9     printf("Enter the Number of elements: ");
10    scanf("%d",&n);
11
12    printf("Enter value of %d elements ",n);
13    printf("\n");
14    for(i=1 ; i<=n;i++){
15        scanf("%d",&arr1[i]);
16    }
17
18    for(i=1 ; i<=n ; i++){
19        sum += arr1[i];
20    }
21    printf("The Sum of the Number is %d",sum);
22}
Process exited after 8.14 seconds with return value 28
Press any key to continue . . .
35°C Mostly cloudy 10:47 AM 24-06-2026
```

Project 3 — Task 4.3

WAP in C for the addition of two matrices

Source Code

```
// Task 4.3 WAP in C to addition of two Matrix

#include<stdio.h>

void main(){
    int a[10][10], b[10][10], sum[10][10];
    int rows, cols, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &rows, &cols);

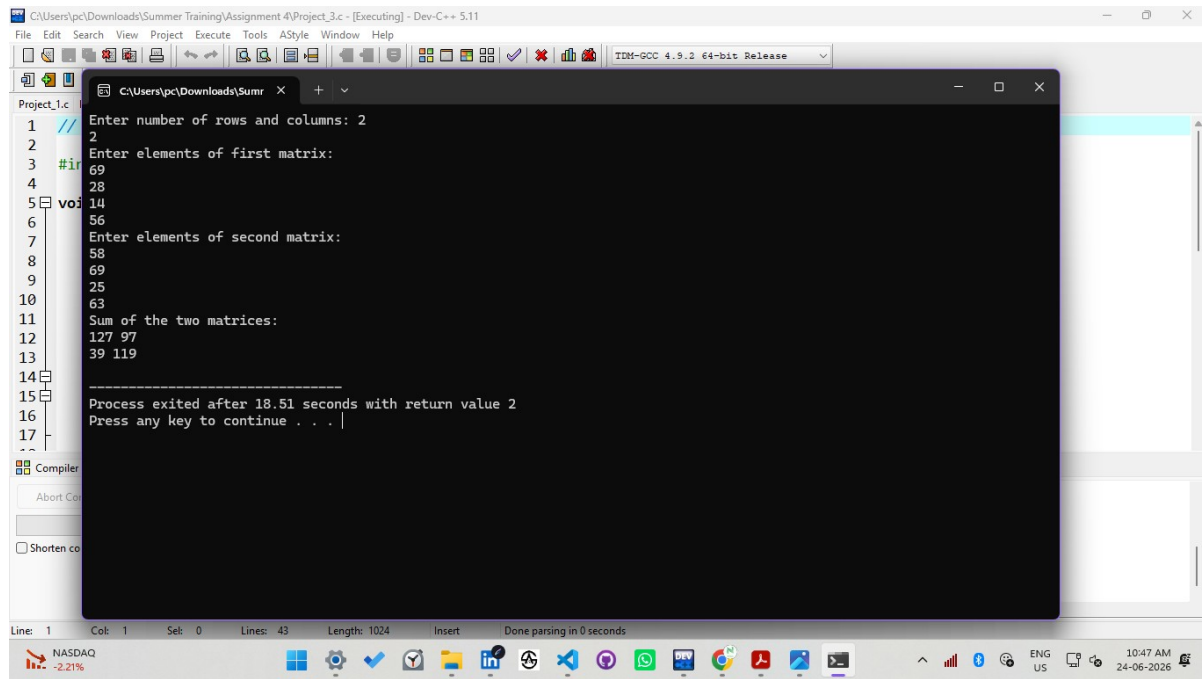
    // Input first matrix
    printf("Enter elements of first matrix:\n");
    for(i = 0; i < rows; i++){
        for(j = 0; j < cols; j++){
            scanf("%d", &a[i][j]);
        }
    }

    // Input second matrix
    printf("Enter elements of second matrix:\n");
    for(i = 0; i < rows; i++){
        for(j = 0; j < cols; j++){
            scanf("%d", &b[i][j]);
        }
    }

    // Add the matrices
    for(i = 0; i < rows; i++){
        for(j = 0; j < cols; j++){
            sum[i][j] = a[i][j] + b[i][j];
        }
    }

    // Display result
    printf("Sum of the two matrices:\n");
    for(i = 0; i < rows; i++){
        for(j = 0; j < cols; j++){
            printf("%d ", sum[i][j]);
        }
        printf("\n");
    }
}
```

Output



```
C:\Users\pc\Downloads\Summer Training\Assignment 4\Project_3.c - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
C:\Users\pc\Downloads\Sumr
Project_1.c
1 // Enter number of rows and columns: 2
2 2
3 Enter elements of first matrix:
4 #11
5 69
6 28
7 14
8 56
9 Enter elements of second matrix:
10 58
11 69
12 25
13 63
14 Sum of the two matrices:
15 127 97
16 39 119
17
-----
Process exited after 18.51 seconds with return value 2
Press any key to continue . . . |
Compiler
Abort Co
Shorten co
Line: 1 Col: 1 Sel: 0 Lines: 43 Length: 1024 Insert Done parsing in 0 seconds
NASDAQ -2.21%
10:47 AM 24-06-2026
```

Project 4 — Task 4.4

WAP in C for the subtraction of two matrices

Source Code

```
// Task 4.4 Substraction of two matrix

#include<stdio.h>

void main(){
    int a[10][10], b[10][10], sum[10][10];
    int rows, cols, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &rows, &cols);

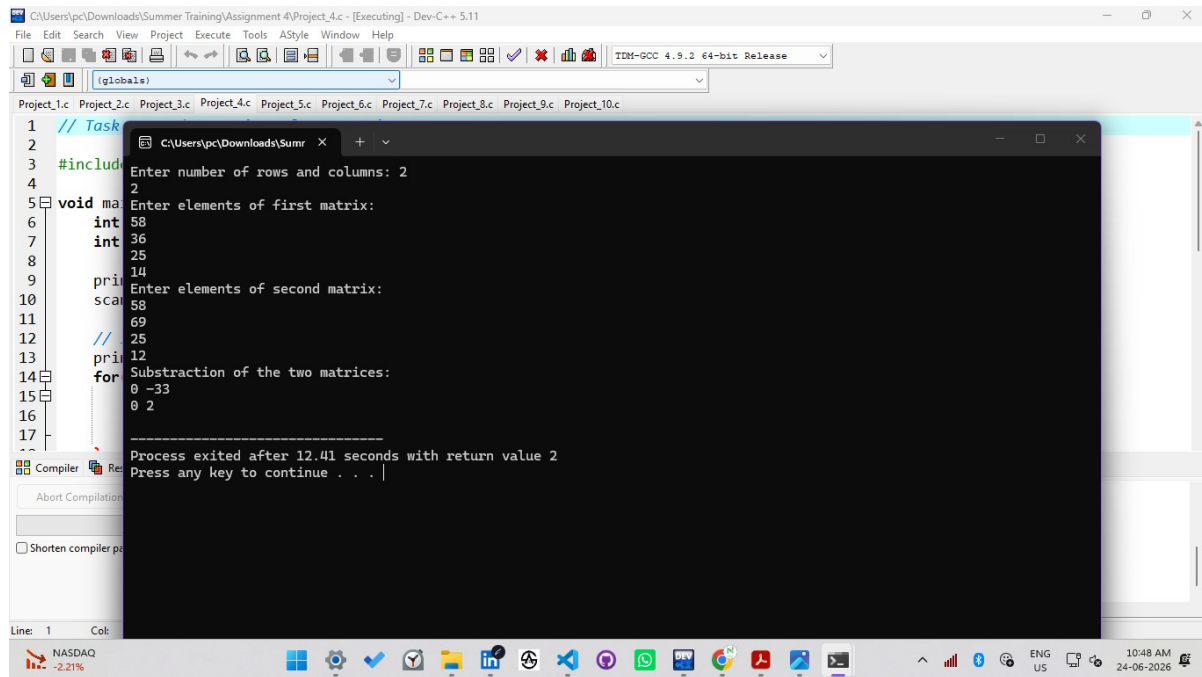
    // Input first matrix
    printf("Enter elements of first matrix:\n");
    for(i = 0; i < rows; i++){
        for(j = 0; j < cols; j++){
            scanf("%d", &a[i][j]);
        }
    }

    // Input second matrix
    printf("Enter elements of second matrix:\n");
    for(i = 0; i < rows; i++){
        for(j = 0; j < cols; j++){
            scanf("%d", &b[i][j]);
        }
    }

    // Add the matrices
    for(i = 0; i < rows; i++){
        for(j = 0; j < cols; j++){
            sum[i][j] = a[i][j] - b[i][j];
        }
    }

    // Display result
    printf("Substraction of the two matrices:\n");
    for(i = 0; i < rows; i++){
        for(j = 0; j < cols; j++){
            printf("%d ", sum[i][j]);
        }
        printf("\n");
    }
}
```

Output



```
1 // Task
2
3 #include <iostream>
4
5 void main()
6 {
7     int rows, cols;
8     int i, j;
9     int a[10][10], b[10][10], c[10][10];
10    Enter number of rows and columns: 2
11    Enter elements of first matrix:
12    58
13    36
14    25
15    14
16    Enter elements of second matrix:
17    58
18    69
19    25
20    12
21    Subtraction of the two matrices:
22    0 -33
23    0 2
24
25    -----
26    Process exited after 12.41 seconds with return value 2
27    Press any key to continue . . . |
```

Compiler: g++
Compiler options: -std=c++11 -O2 -fcommon -fPIE -pie -fstack-protector-strong -fstack-protector -fstack-clash-protection -fcf-protection -fdebug-prefix-map=/usr/src/debug-10.1.0-1.fc35 -fstack-protector-strong -fstack-protector -fstack-clash-protection -fcf-protection -fdebug-prefix-map=/usr/src/debug-10.1.0-1.fc35

Line: 1 Col: 1

NASDAQ -2.21%

10:48 AM 24-06-2026

Project 5 — Task 4.5

WAP in C to find the transpose of a matrix

Source Code

// Task 4.5 WAP in C to find the Transpose of a Matrix

```
#include<stdio.h>

int main(){
    int a[10][10], transpose[10][10];
    int rows, cols, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &rows, &cols);

    // Input matrix
    printf("Enter elements of the matrix:\n");
    for(i = 0; i < rows; i++){
        for(j = 0; j < cols; j++){
            scanf("%d", &a[i][j]);
        }
    }

    // Find transpose
    for(i = 0; i < rows; i++){
        for(j = 0; j < cols; j++){
            transpose[j][i] = a[i][j];
        }
    }

    // Display original matrix
    printf("Original matrix:\n");
    for(i = 0; i < rows; i++){
        for(j = 0; j < cols; j++){
            printf("%d ", a[i][j]);
        }
        printf("\n");
    }

    // Display transposed matrix
    printf("Transpose of matrix:\n");
    for(i = 0; i < cols; i++){
        for(j = 0; j < rows; j++){
            printf("%d ", transpose[i][j]);
        }
        printf("\n");
    }
}
```

Output

```
CAUsers\pc\Downloads\Summer Training\Assignment 4\Project_5.c - [Executing] - Dev-C++ 5.11
Enter number of rows and columns: 2
2
Enter elements of the matrix:
36
69
58
47
Original matrix:
36 69
58 47
Transpose of matrix:
36 58
69 47

=====
Process exited after 10.81 seconds with return value 2
Press any key to continue . . .
```

Line: 1 Col: 1 Sel: 0 Lines: 44 Length: 1037 Insert Done parsing in 0 seconds

Hot days ahead 35°C

10:48 AM 24-06-2026

Project 6 — Task 4.6

WAP in C to illustrate the use of the `strcat()` function

Source Code

```
// Task 4.6 WAP in C to illustrate the use of strcat() function

#include<stdio.h>
#include<string.h>

void main(){
    char str1[50], str2[30];

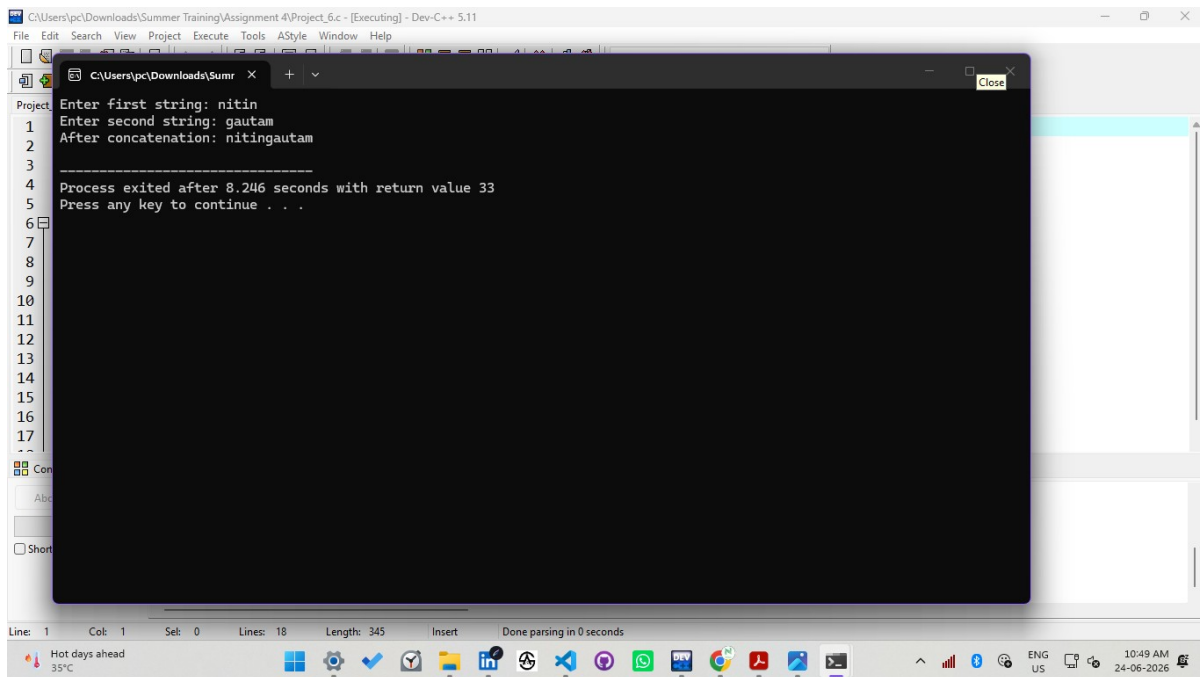
    printf("Enter first string: ");
    gets(str1);

    printf("Enter second string: ");
    gets(str2);

    strcat(str1, str2);

    printf("After concatenation: %s\n", str1);
}
```

Output



The screenshot shows a Dev-C++ IDE window titled "C:\Users\pc\Downloads\Summer Training\Assignment 4\Project_6.c - [Executing] - Dev-C++ 5.11". The IDE has a menu bar (File, Edit, Search, View, Project, Execute, Tools, AStyle, Window, Help) and a toolbar. A console window is open, displaying the following output:

```
Enter first string: nitin
Enter second string: gautam
After concatenation: nitingautam

-----
Process exited after 8.246 seconds with return value 33
Press any key to continue . . .
```

The console window has a "Close" button in the top right corner. The IDE's status bar at the bottom shows "Line: 1", "Col: 1", "Sel: 0", "Lines: 18", "Length: 345", "Insert", "Done parsing in 0 seconds". The Windows taskbar at the very bottom shows the date and time as "10:49 AM 24-06-2026" and the language as "ENG US".

Project 7 — Task 4.7

WAP in C to illustrate the use of the strlen() function

Source Code

```
// Task 4.7 WAP in c to illustrate the use of strlen() function.

#include<stdio.h>
#include<string.h>

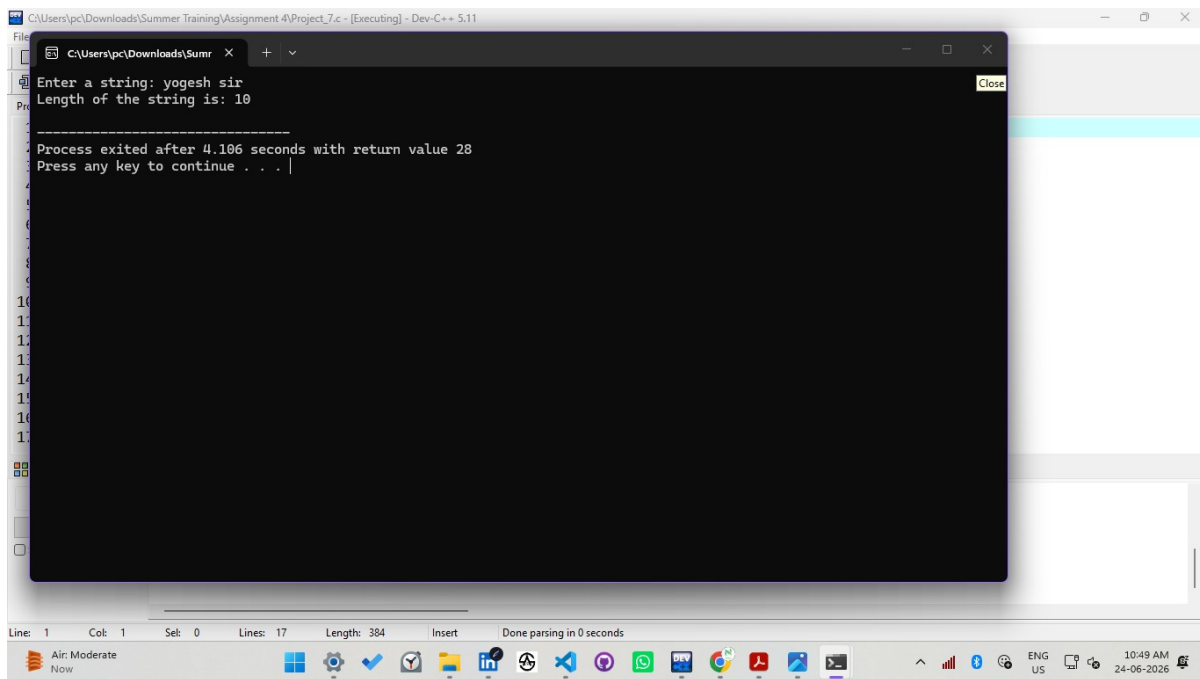
void main(){
    char str[50];
    int length;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
    str[strcspn(str, "\n")] = '\0'; // remove trailing newline

    length = strlen(str);

    printf("Length of the string is: %d\n", length);
}
```

Output



```
C:\Users\pc\Downloads\Summer Training\Assignment 4\Project_7.c - [Executing] - Dev-C++ 5.11
Enter a string: yogesh sir
Length of the string is: 10

Process exited after 4.106 seconds with return value 28
Press any key to continue . . . |
```

Project 8 — Task 4.8

WAP in C to illustrate the use of the strcpy() function

Source Code

```
// Task 4.8 WAP in c to illustrate the use of strcpy() function.

#include<stdio.h>
#include<string.h>

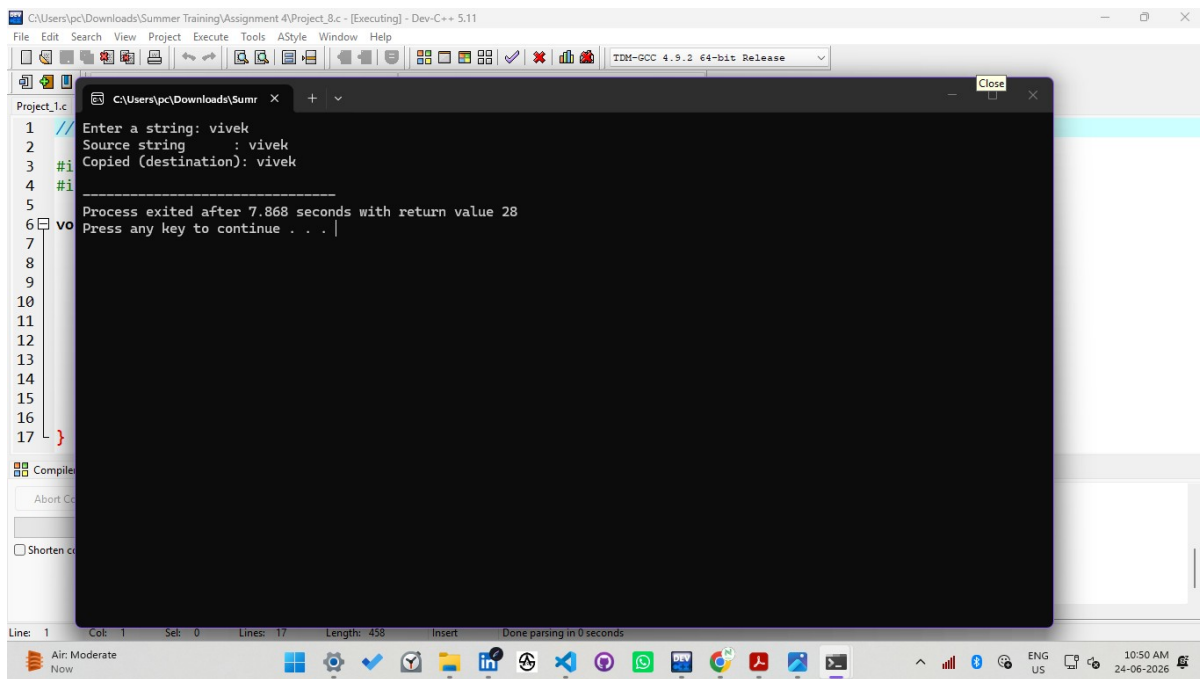
void main(){
    char source[50], destination[50];

    printf("Enter a string: ");
    fgets(source, sizeof(source), stdin);
    source[strcspn(source, "\n")] = '\0'; // remove trailing newline

    strcpy(destination, source);

    printf("Source string      : %s\n", source);
    printf("Copied (destination): %s\n", destination);
}
```

Output



The screenshot shows the Dev-C++ IDE with a project named 'Project_1.c'. The code is as follows:

```
1 // Enter a string: vivek
2 Source string      : vivek
3 Copied (destination): vivek
4 #include<stdio.h>
5 #include<string.h>
6
7 void main()
8 {
9     char source[50], destination[50];
10
11     printf("Enter a string: ");
12     fgets(source, sizeof(source), stdin);
13     source[strcspn(source, "\n")] = '\0'; // remove trailing newline
14
15     strcpy(destination, source);
16
17     printf("Source string      : %s\n", source);
18     printf("Copied (destination): %s\n", destination);
19 }
```

The output window shows the following execution details:

```
Process exited after 7.868 seconds with return value 28
Press any key to continue . . .
```

The status bar at the bottom indicates 'Line: 1', 'Col: 1', 'Sel: 0', 'Lines: 17', 'Length: 458', 'Insert', 'Done parsing in 0 seconds', and the system clock shows '10:50 AM 24-06-2026'.

Project 9 — Task 4.9

WAP in C to illustrate the use of the strcmp() function

Source Code

```
// Task 4.9 WAP in C to illustrate the use of strcmp() function.
```

```
#include<stdio.h>
#include<string.h>

void main(){
    char str1[50], str2[50];
    int result;

    printf("Enter first string: ");
    fgets(str1, sizeof(str1), stdin);
    str1[strcspn(str1, "\n")] = '\0';

    printf("Enter second string: ");
    fgets(str2, sizeof(str2), stdin);
    str2[strcspn(str2, "\n")] = '\0';

    result = strcmp(str1, str2);

    if(result == 0){
        printf("Both strings are equal.\n");
    }
    else if(result < 0){
        printf("String 1 is less than String 2.\n");
    }
    else{
        printf("String 1 is greater than String 2.\n");
    }

    printf("strcmp() returned: %d\n", result);
}
```

Output

The screenshot displays the Dev-C++ IDE interface. The main window shows a C++ source file with the following code:

```
1 // Task
2
3 #include <string.h>
4 #include <conio.h>
5
6 void main()
7 {
8     char str1[100];
9     int i;
10    printf("Enter first string: ");
11    fgets(str1, 100, stdin);
12    str1[strcspn(str1, "\n")] = '\0';
13
14    printf("Enter second string: ");
15    fgets(str2, 100, stdin);
16    str2[strcspn(str2, "\n")] = '\0';
17
18    printf("String 1 is less than String 2.\n");
19    printf("strcmp() returned: %d\n", strcmp(str1, str2));
20
21    printf("-----\n");
22    printf("Process exited after 27.9 seconds with return value 22\n");
23    printf("Press any key to continue . . . ");
24    getch();
25}
```

The output window shows the following text:

```
Enter first string: harsh
Enter second string: lawaniya
String 1 is less than String 2.
strcmp() returned: -1
-----
Process exited after 27.9 seconds with return value 22
Press any key to continue . . .
```

The IDE status bar at the bottom indicates the current line and column, and the system tray shows the time as 10:50 AM on 24-06-2026.

Project 10 — Task 4.10

WAP in C to illustrate the use of the `strrev()` function

Source Code

```
// Task 4.10 WAP in c to illustrate the use of strrev() function.

#include<stdio.h>
#include<string.h>

void main(){
    char str[50];

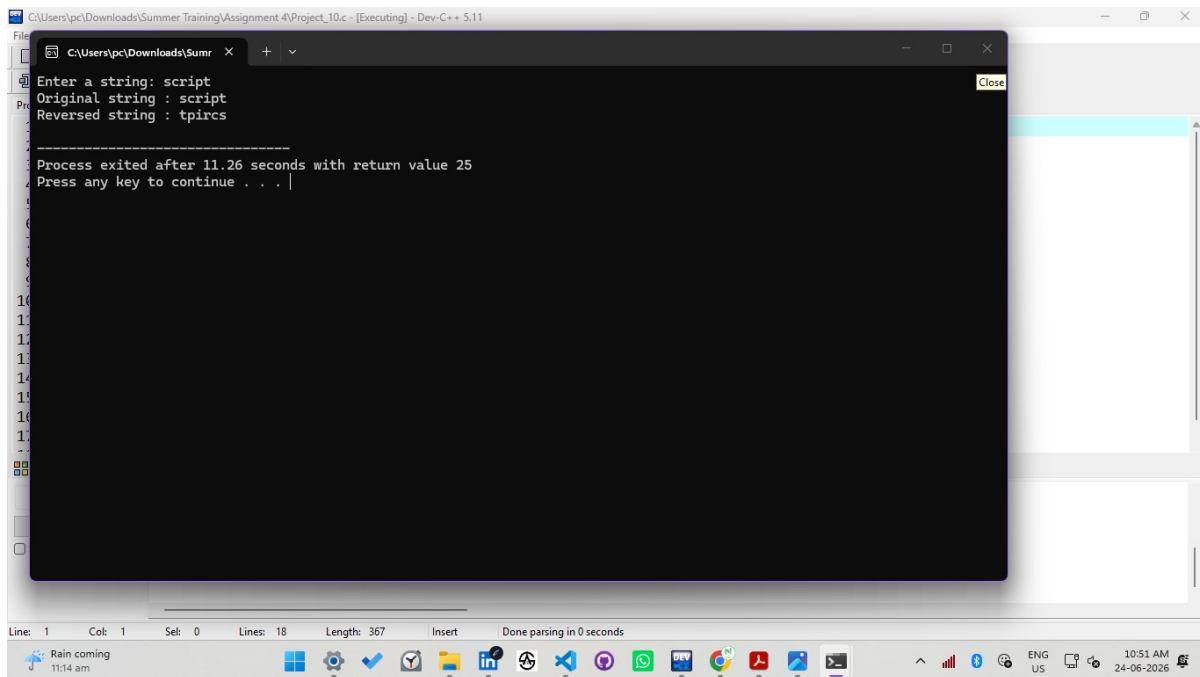
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
    str[strcspn(str, "\n")] = '\0';

    printf("Original string : %s\n", str);

    strrev(str);

    printf("Reversed string : %s\n", str);
}
```

Output



```
C:\Users\pc\Downloads\Summer Training\Assignment 4\Project_10.c - [Executing] - Dev-C++ 5.11
Enter a string: script
Original string : script
Reversed string : tpirsc

Process exited after 11.26 seconds with return value 25
Press any key to continue . . .
```